

Audience-dependent AI explanations in finance

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UNIVERSITÀ DEGLI STUDI DI NAPOLI
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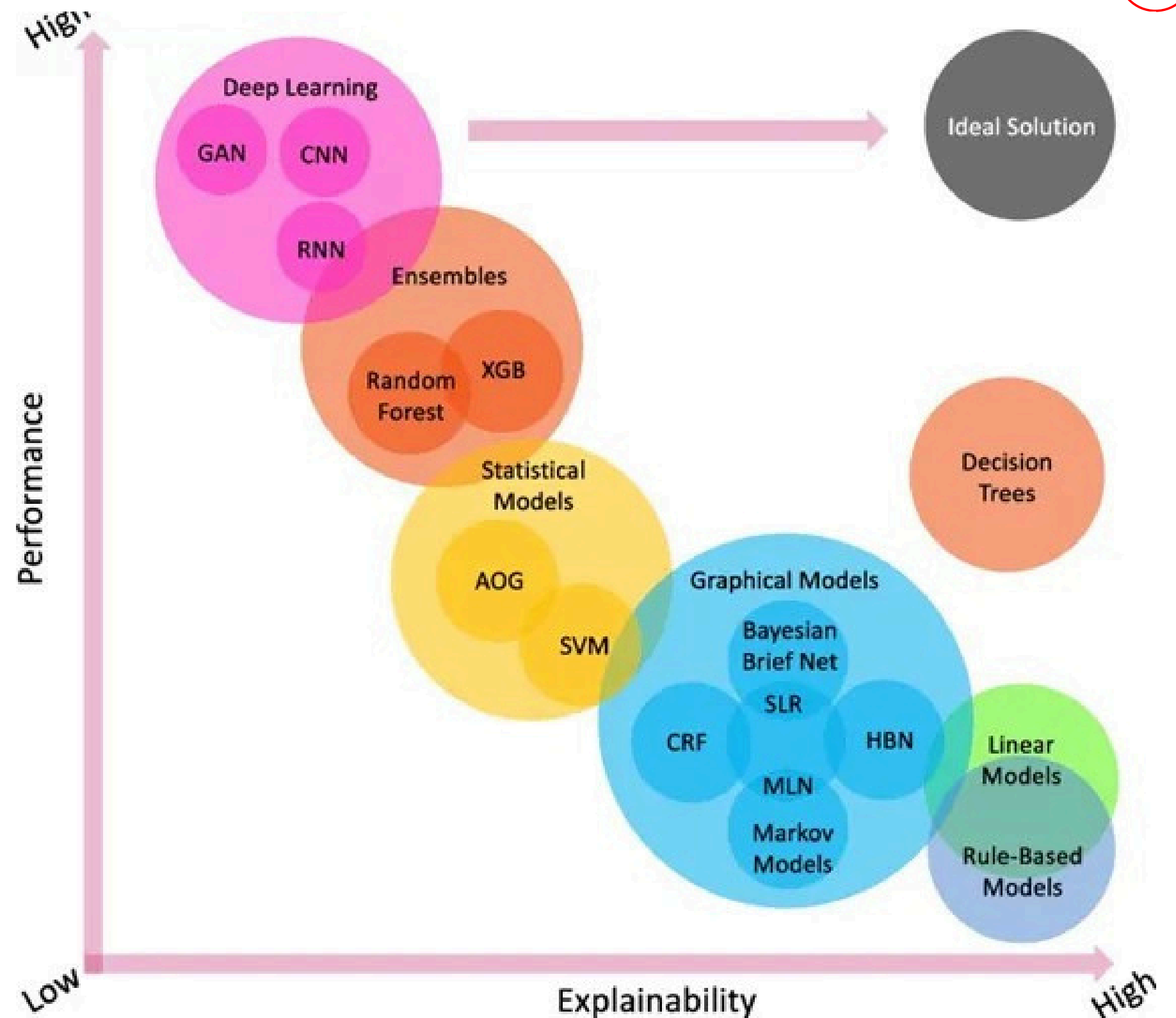
Key thematic areas: **The European Financial Digital Space, AI-driven analysis of financial markets, Explainable and fair AI in finance, Blockchain infrastructure and innovation, Sustainability and green finance**



What are blackbox models?

MODELS WHOSE INTERNAL LOGIC IS NOT DIRECTLY INTERPRETABLE BY HUMANS

- "White-box": linear regression, decision trees vs "black-box" neural networks or ensemble methods
- Literature in finance (see Zetzsche et al., 2020; Kumar, 2017; Cernevicene et al., 2024; Ogunniyi, et al., 2025)



Why choose black– boxes?

"THE CURSE & GIFT OF DEPTH"



Optimization

Models learn by minimizing the loss through gradient descent.



Non-linearity

Activation functions (e.g., ReLU, sigmoid) allow complex decision boundaries.



Overparameterization and emergent structure

Enables pattern discovery but hides reasoning

Deep layers encode abstract manifolds = statistically efficient, but cognitively opaque

Explainable AI

METHODS THAT MAKE BEHAVIOR, DECISIONS,
AND INTERNAL LOGIC UNDERSTANDABLE
TO HUMANS



Scope

Global - understand the overall behavior of the model

e.g., feature importance, partial dependence plots (PDP)

Local - explain a single prediction
e.g., LIME, SHAP, counterfactuals

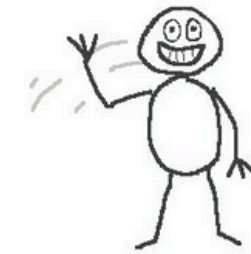


Dependence

Model-specific - works with only certain algorithms (e.g., coefficients in linear / logistic regression, tree visualizations)

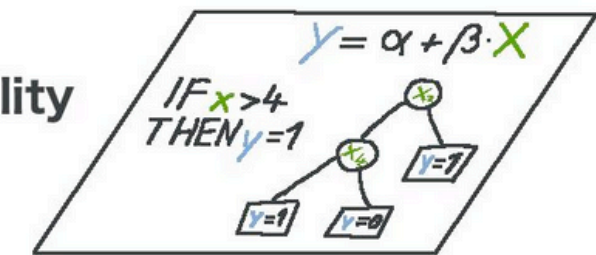
Model-agnostic - works with any model by proving input-output behavior (e.g., SHAP, LIME)

Humans



↑ inform

Interpretability
Methods



↑ extract

Black Box
Model



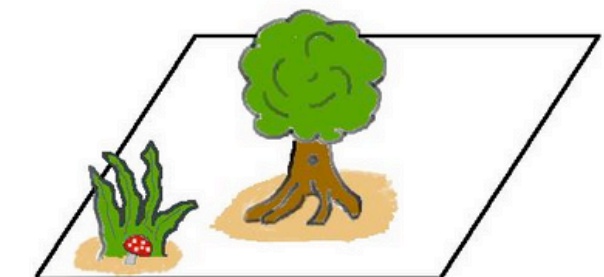
↑ learn

Data

X ₁	X ₂	X ₃	...	X _n
1.5	2.1	0.8	...	0.5
2.3	1.7	1.2	...	1.1
0.9	3.4	0.5	...	0.2
...	...	...	...	...
...	...	...	...	...
...	...	...	...	...
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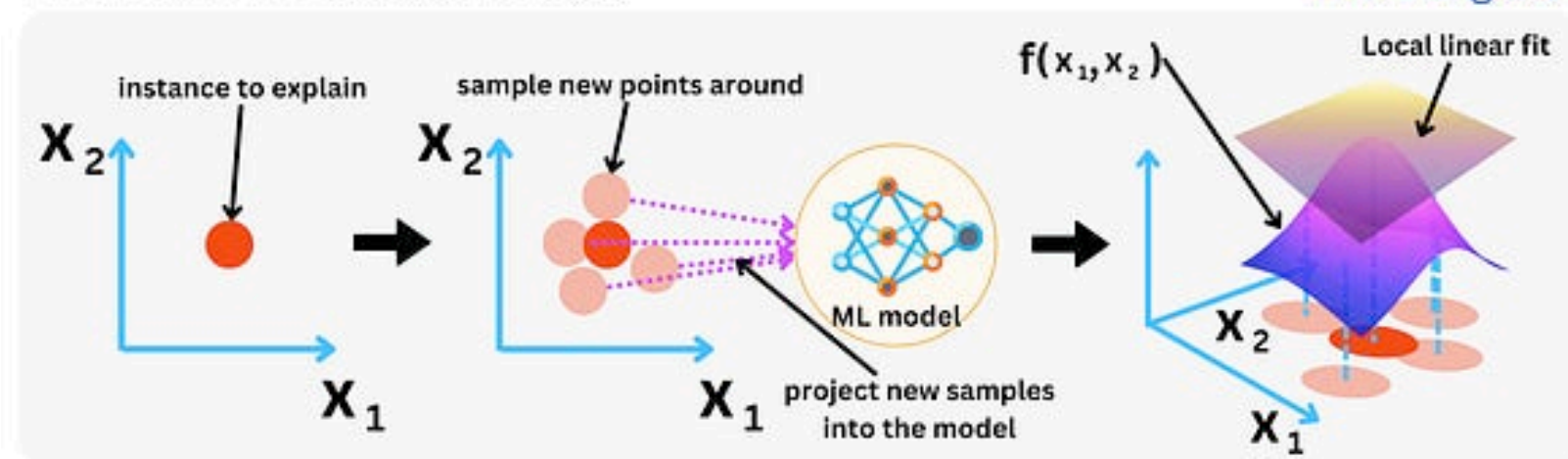
↑ capture

World

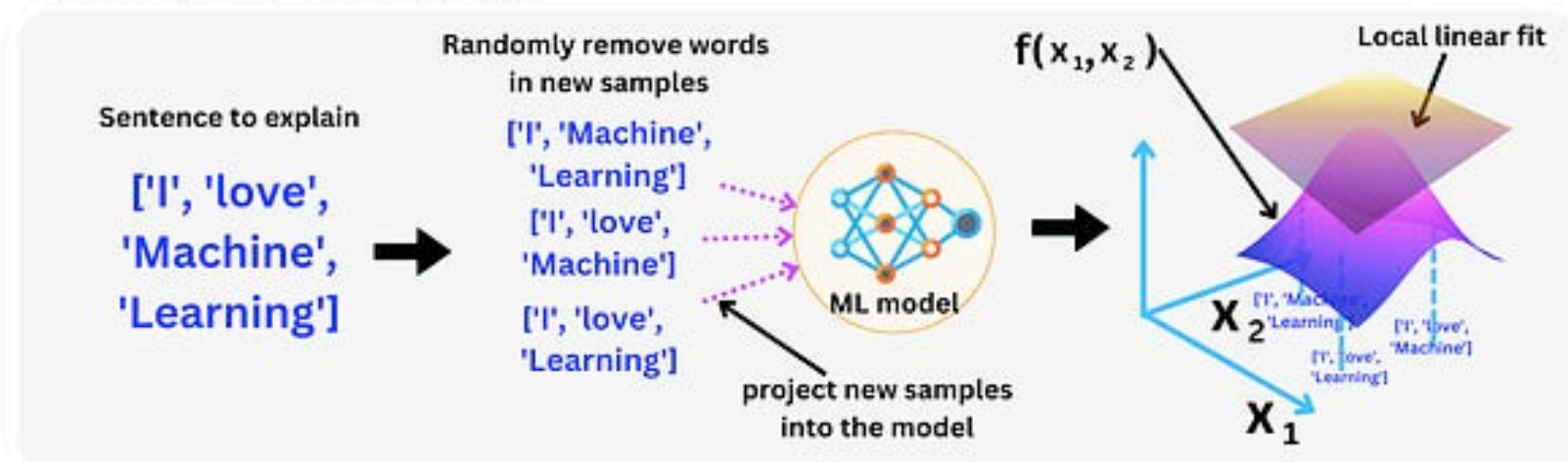


Explainable AI: LIME

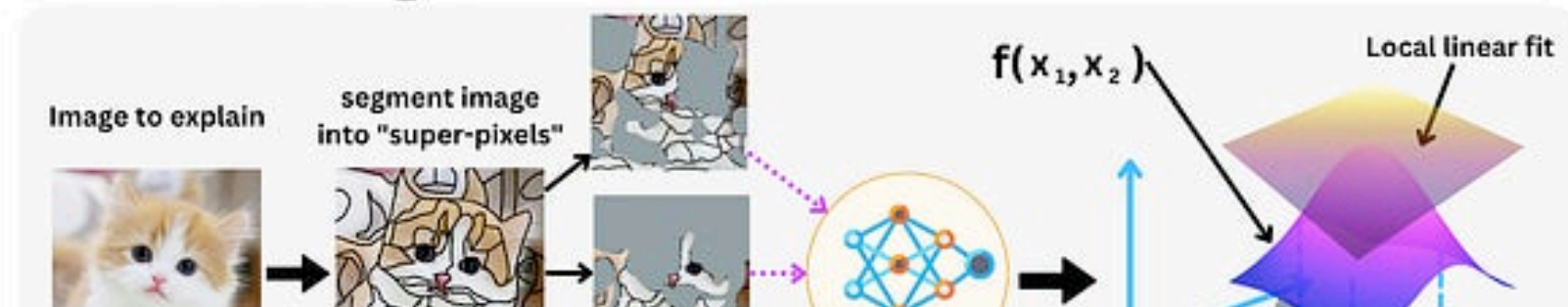
LIME with Tabular data



LIME with Text data



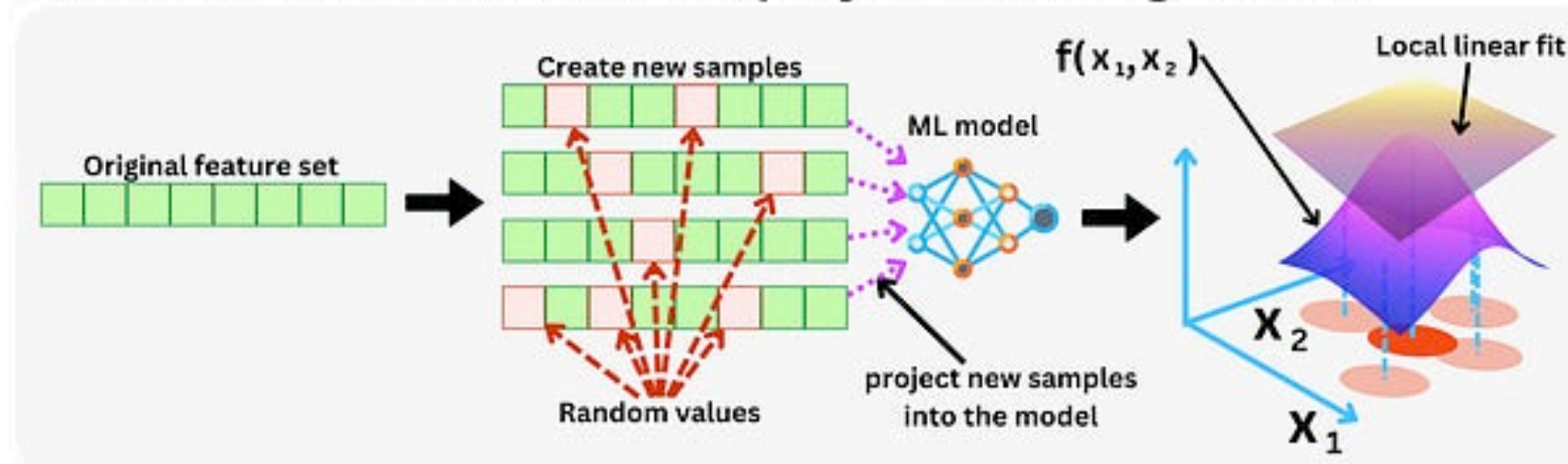
LIME with Image data



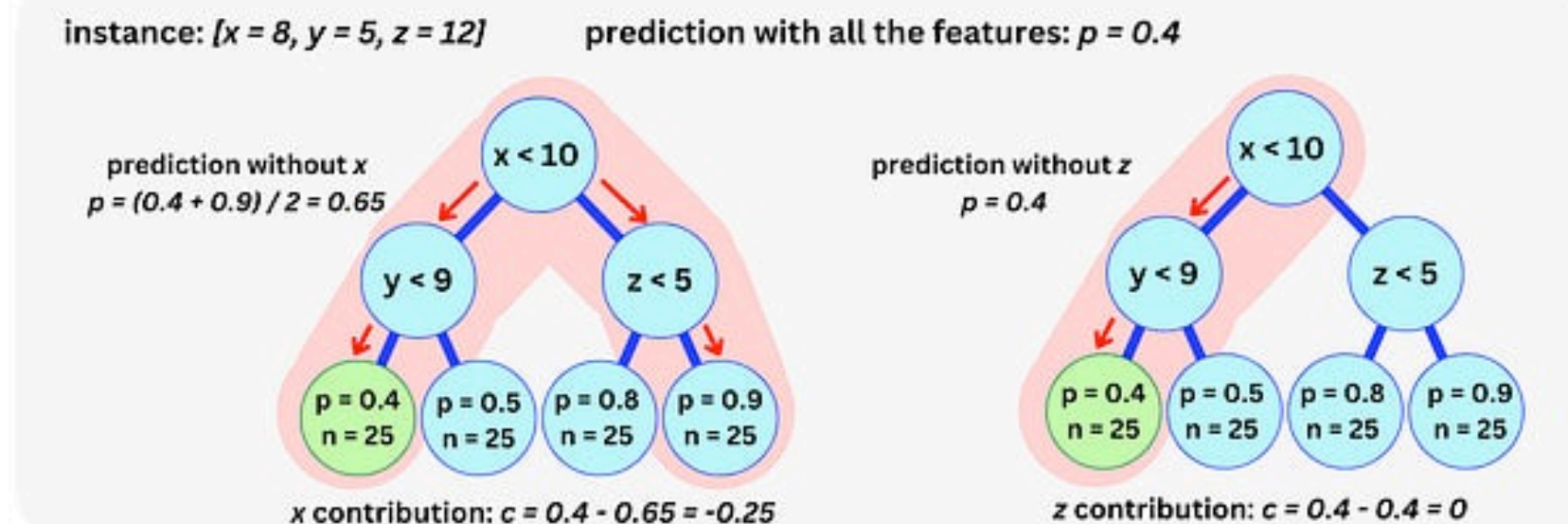
Explainable AI: SHAP

TheAiEdge.io

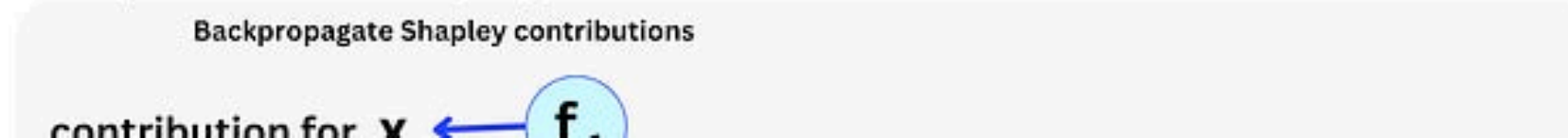
Kernel SHAP: LIME with Shapley Smoothing Kernel



Tree SHAP: Shapley estimates for Trees



Deep SHAP: Shapley estimates for Neural Networks

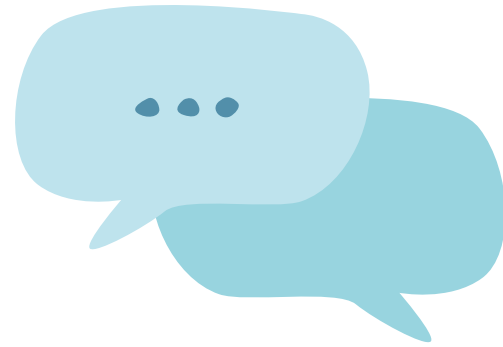


Exercise on SHAP: How toxic (are you)?

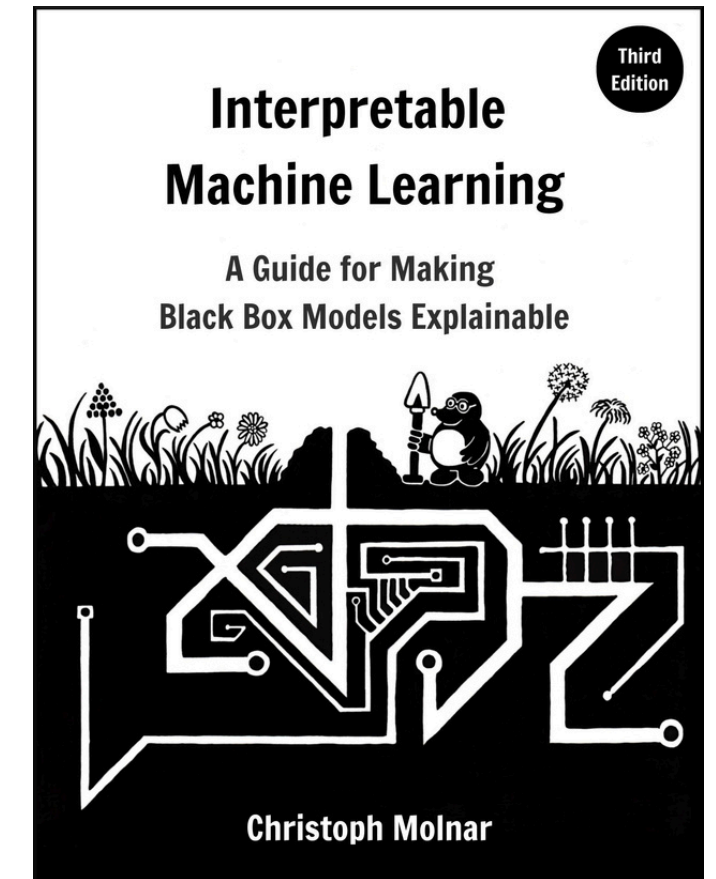
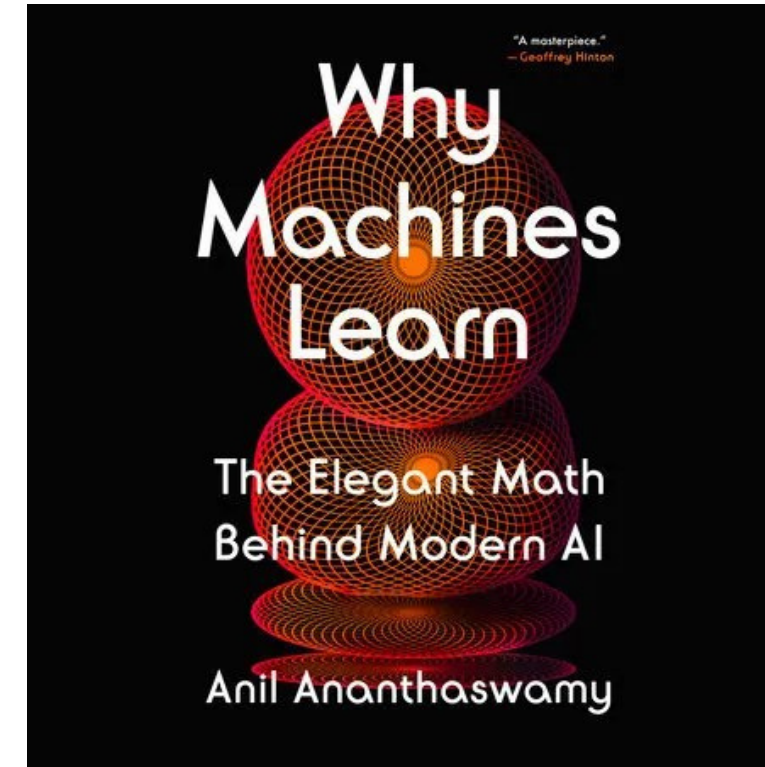
Type any short text - it could be a social media post, a product review, or even something from news or song - and click "COMPUTE SHAP VALUES"

- Guide questions
 - Which words were marked red or blue?
 - Were you surprised by what was considered toxic or negative?
 - What does this tell us about explainability methods like shap?





Further reading



- Ananthaswamy, Anil, Why Machines Learn: The Elegant Math Behind Modern AI. Dutton, 2024.
- Molnar, C. (2022). Interpretable Machine Learning.

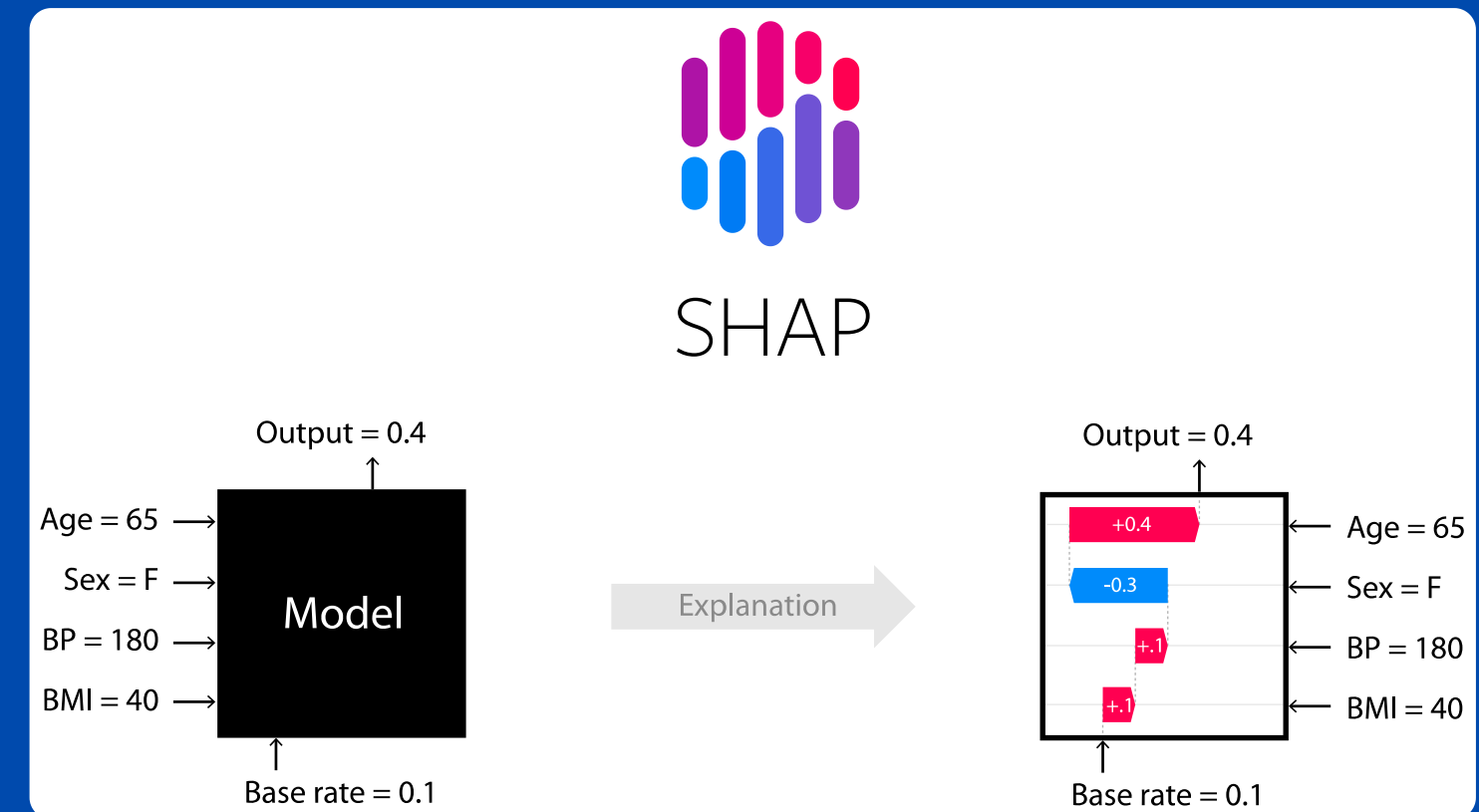
Research Problem

WHAT I WANT TO SOLVE

XAI methods in finance are focused on technical users (SHAP, LIME) like "inmates running the asylum" (Miller et al., 2017)

Incomprehensible AI leads to low AI trust and adoption (Bernardo & Seva, 2023)
Overtrust in incapable AI (Ehsan et al., 2021)

Literature suggests audience-dependent explanations (Misheva & Osterrieder, 2023; Schmude et al., 2024)

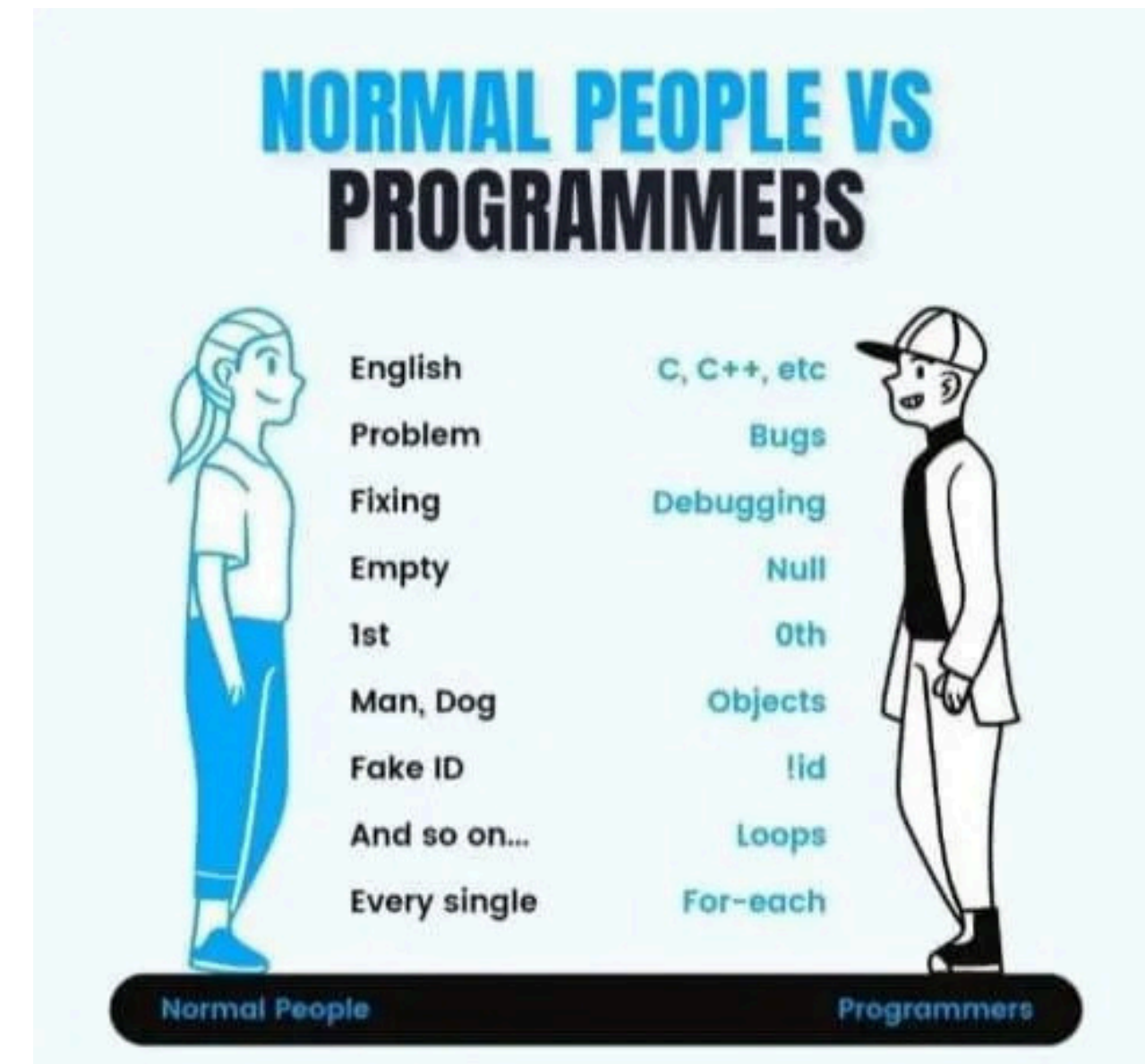


Objective

HOW I WANT TO SOLVE IT

Map the **explainability requirements** of different roles (Bracke, 2019; Molnar, 2025) via mixed-method data analysis

1. Method: Exploratory semi-structured interviews and quantitative survey (English and Italian) to test how preferences cluster statistically
2. Statistical analysis: Item Response Theory, Cluster analysis



Role	Responsibility
Developer	Build or train the model
Operator	Run or monitor the system
Executor	Decide based on outputs
Validator	Investigate the system's result
Manager	Responsible for application
Conduct Regulator	Ensure adherence to customer protection, fair
Prudential Regulator	Ensure adherence to capital, risk, systematic

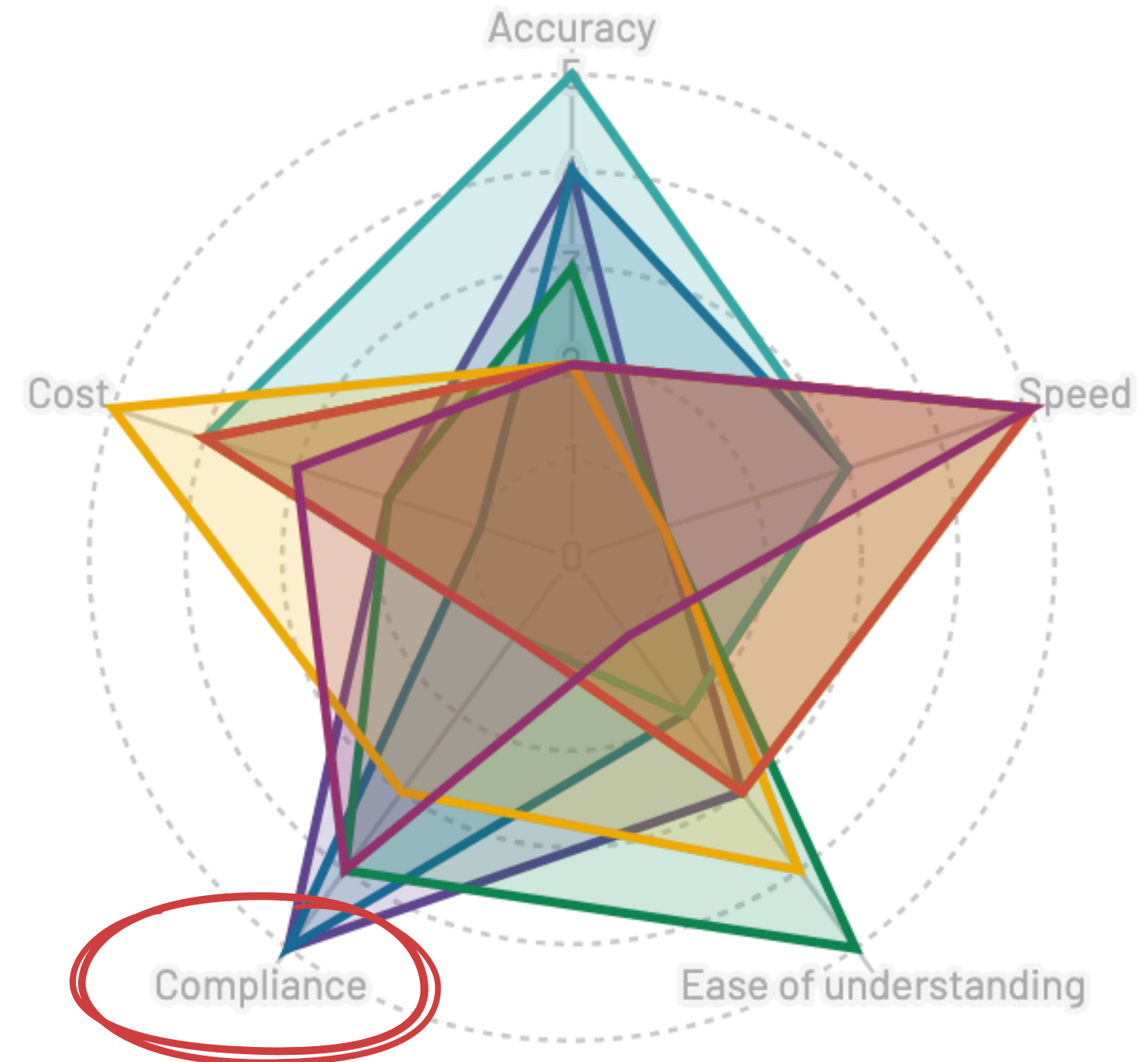
What we know so far

FROM FINANCE PROFESSIONAL INTERVIEWS

1. Explanations must have **evidence** for compliance and support auditability.
2. Business users want **simple** and usable reasons; innovators and risk staff need traceability to data, policies and **metrics**.
3. **Accuracy and compliance** outweighs speed and cost (efficiency gains).
4. Accuracy varies per role, from faithfulness to source data to reducing false positives and errors.

Legend for roles in the radar chart:

- Credit risk modeller (Purple)
- Corporate banker (Blue)
- Product manager (Teal)
- Branch manager (Green)
- Research (Light Green)
- Data scientist (Yellow)
- Risk manager (Orange)
- Risk monitoring (Red)
- CEO (Dark Purple)



Next steps

Launch survey to financial
professionals

Constructs: Qualitative aspects of XAI
(unobservable), hence "latent". E.g., explanation
needs, experiences with AI, perceived qualitym
preferences

**Suitable for Item Response Theory, EFA, or
Cluster analysis**

Open issues in the survey: results from short pilot test

- Which scale to use? 4 or 5-scale?
- Long & boring, but trade-off with data quality and recommender system needs

Let me know your feedback through the end of each section.

Grazie mille!
Forza Napoli



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